Single-Strand Conformation Polymorphism Analysis for Identification of Four Members of the *Anopheles funestus* (Diptera: Culicidae) Group

LIZETTE L. KOEKEMOER,[1] LAURENCE LOCHOUARN,[2] RICHARD H. HUNT,[1] AND MAUREEN COETZEE[1]

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ABSTRACT Members of the *Anopheles funestus* Giles group are difficult to identify because of the morphological overlap that exists within the group. This inability to distinguish species, as well as the fact that the species vary in their behavior and biting preferences, complicate the successful planning and maintaining of malaria control programs. In this article we discuss the use of a single-strand conformation polymorphism (SSCP) assay to distinguish 4 members of the *An. funestus* group collected at 10 different localities in Africa. rDNA genes differ at numerous sites among closely related species. Using conserved primers, the D3 domain in the 28S gene was amplified, electrophoresed on SSCP gels, and species-specific patterns were observed. Intraspecific variation was detected in *An. funestus* specimens from East and West Africa. Analyzing 108 *An. funestus*, 78 *An. vaneedeni* Gillies & Coetzee, 21 *An. rivulorum* Leeson, and 2 *An. leesoni* Evans, we concluded that SSCPs can be used successfully as a molecular tool for the identification of these species.

KEY WORDS *Anopheles funestus* group, identification, single-strand conformation polymorphism analysis

SPECIES COMPLEXES HAVE been shown to exist in several groups of medically important Diptera. Complexes are groups of species that are similar morphologically but exhibit behavioral and vectorial differences. One problem in vector control today is the difficulty of identifying the cryptic species within these groups or complexes. This identification is necessary because not all the species within these groups are involved equally in the transmission of pathogens to humans. A classical example is the *Anopheles gambiae* Giles complex, where only 2 of a total of 7 recognized species are major vectors of the malaria parasite *Plasmodium falciparum* (Gillies and deMeillon 1968, Service 1985, Gillies and Coetzee 1987, Hunt et al. 1998). Rapid large-scale identification of the species in this complex was greatly improved by the development of the species-specific polymerase chain reaction (PCR) assay (Scott et al. 1993). This technique is routinely used in many laboratories to identify members of the *An. gambiae* complex (Paskewitz et al. 1993, Van Rensburg et al. 1997, Temu et al. 1998, Toure et al. 1998). The 3rd major vector species of malaria in sub-Saharan Africa belongs to another group of species, the *Anopheles funestus* Giles group.

The *An. funestus* group consists of at least 8 species—the nominal vector *An. funestus*, *An. vaneedeni* Gillies & Coetzee, *An. rivulorum* Leeson, *An. leesoni* Evans, *An. confusus* Evans & Leeson, *An. parensis* Gillies, *An. brucei* Service, and *An. aruni* Sobti. *An. funestus* has been recognized as a major vector of malaria parasites since the early part of the 20th century (Gillies and De Meillon 1968). The only other member of the group that has been found to be infected naturally with *P. falciparum* is *An. rivulorum* (Wilkes et al. 1996). This species is widespread in Africa and is usually zoophilic. However, if cattle are scarce or not available, this species will feed on humans and can be infected naturally with *P. falciparum*. *Anopheles vaneedeni*, occurring in southern Africa, has been infected with *P. falciparum* in the laboratory but has not been found naturally infected with the parasite (De Meillon et al. 1977, Gillies and Coetzee 1987).

The major problem when working with the *An. funestus* group is the difficulty in identifying the different species. Morphological characteristics of the adults are variable and overlap in many instances. *An.vaneedeni* and *An. parensis* can be separated from *An. funestus* by, if present, pale scaling on the tarsomeres and male palps respectively (Gillies and Coetzee 1987). Larval morphology can be used to identify *An. confusus* and *An. rivulorum*, and *An. leesoni* can be identified on the basis of both larval and egg morphology (Gillies and De Meillon 1968, Gillies and Coetzee 1987). Polytene chromosome banding arrangements can distinguish *An. funestus*, *An. rivulorum*, *An. leesoni*, *An. parensis*, and *An. confusus* from one another

[1] Medical Entomology, Department of Tropical Diseases, School of Pathology of the South African Institute for Medical Research and the University of the Witwatersrand, P.O. Box 1038, Johannesburg, 2000 South Africa.

[2] Laboratoire de Zoologie Médicale, Institut de Recherche Scientifique pour le Developpement en Cooperation (ORSTOM), Institut Pasteur, B.P. 220, Dakar, Sénégal.

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(Green 1982). However, *An. funestus* and *An. vaneedeni* have homosequential banding patterns and differ by a single floating inversion on arm 2 (Green and Hunt 1980). Polytene chromosomes are found in the ovaries of half-gravid females or the salivary glands of the 4th instars, and the technique needs a great deal of expertise to analyze the results. Morphological identification however, needs all the life stages of the mosquito (from the eggs to the adult). This method is time-consuming and mortality is high when rearing isofemale lines under laboratory conditions.

Molecular techniques have the advantage of being rapid and inexpensive, applicable to all life stages and, for a PCR assay, limited material is needed. To date only one molecular DNA-based method exists to identify members of the *An. funestus* group, that being a restriction fragment-length polymorphism (RFLP) assay for distinguishing *An. funestus* from *An. vaneedeni* (Koekemoer et al. 1998).

Single-strand conformation polymorphism (SSCP) is a technique that has been used recently as a molecular taxonomic tool. Hayashi (1991) gave a good overview of the practical use of SSCP electrophoresis and found that mutated clones with the same SSCP banding pattern have the same sequence mutations. Hiss et al. (1994) used SSCP to show that species-diagnostic banding patterns could be achieved for various taxa by this relatively new technique.

The length, location, and number of intrastrand base pairs determine the secondary and tertiary structure of a conformation of single-stranded DNA (Orita et al. 1989). SSCP electrophoresis is based on this principal, and the difference in the DNA sequence will alter the conformation of single-strand DNA that will result in different mobilities during electrophoresis. The PCR products are denatured and then directly plunged into ice to help in the formation of intrastrand complexes and reducing the renaturing of complementary strands. Formamide dye and a neutral polyacrylamide gel help to maintain the molecules secondary conformation during electrophoresis (Hiss et al. 1994).

Currently it is not possible to predict the mobility shift because of mutation differences, however shorter fragments (100–400 bp) are more sensitive for the detection of mutation variation (Hayashi 1991). Mutations incorporated following speciation should be readily detected according to Hiss et al. (1994). Electrophoresis conditions (e.g., temperature changes and the concentration of ions and solvents) may influence the sensitivity of mutation detection, and the electrophoresis process must be standardized for each set of conditions. SSCP electrophoresis has been used in different mutation studies such as in the human genetic field for detection of human genetic diseases and genetic markers (Orita 1989) and in phylogenetic studies of wheat and grass (Wang 1997).

Here we describe the use of SSCPs for distinguishing between 4 members of the *An. funestus* group.

Materials and Methods

Field Collections and a priori Identifications. The samples used in this study came from the following 7 countries: (1) Madagascar, Ankazobe (18° 19′ S, 47° 07′ E), caught biting humans indoors April 1996; (2)Tanzania, Bagamoyo (6° 26′ S, 38° 55′ E) at Matimbwa Village resting indoors December 1992; (3) Botswana, Shakawe (18° 10′ S, 21° 43′ E), resting indoors March 1994; (4) South Africa, Komatipoort (25° 26′ S, 31° 57′ E), biting humans outdoors and CO^2-baited net traps, January 1996, 1997; Tzaneen, Thomo Camp, Giyani (23° 15′ S, 30° 47′ E), pit trap December 1997, Mamfene (27° 23′ S, 32° 16′ E), feeding on cattle March 1998; (5) Mozambique, Boane (26° 02′ S, 32° 19' E), caught resting indoors, May 1997; (6) Côte d'Ivoire, Petionara (near Bouake) (7° 42′ N, 5° 0′ W) caught inside houses, June 1997; Abidjan (15°19′ N, 4° 1′ W) caught inside houses, June 1997; (7) Senegal, caught indoors using pyrethrum spray catch: Dielmo (13° 45′ N, 16° 25′ W), February 1998; Kedougou (12° 33′ N, 12° 11′ W) October 1997 and Sankagne (13° 24′ N, 13° 45′ W), November 1997.

All samples, except those from Senegal, were transported to the laboratory where females were held in individual tubes for egg laying. Egg morphology was examined before eggs were put into bowls with aerated water. Larvae were fed on a high-protein diet consisting of ground brewers' yeast tablets and dog biscuits. Larval and pupal pelts were stored initially in 80% ethanol, then permanently mounted on microscope slides to be used with pinned adults for morphological identification. The remainder of the adults emerging from individual egg batches were frozen in liquid nitrogen. Morphological identification was carried out according to Gillies and De Meillon (1968) and Gillies and Coetzee (1987). The Senegal samples were identified chromosomally (Green and Hunt 1980).

DNA Extraction, Amplification, and Sequencing. The primers used in amplification were designed by W. Kelly Thomas for use in nematode studies (R. Sharpe, Leeds University, UK, personal communication). They amplify a variable D3 domain in the 28S gene. Sequences of the primers are as follows:

D3A sense: 5′ GAC CCG TCT TGA AAC ACG GA 3′

D3B antisense: 5′ TCG GAA GGA ACC AGC TAC TA 3′

Amplifications were done using the template from DNA extractions (Collins et al. 1987) or from single legs centrifuged at 16, 000 × *g* for 4 min with the PCR mixture. The PCR mixture contained the following: 2.5 μl of the 10x reaction buffer (500 mM KCl, 100 mM Tris-HCl pH 8.3), 1.5 mM $MgCl_2$, 1 μM of each primer, 200 μM of each dNTP, 2 U thermostable taq DNA polymerase, overlaid by 30 μl of mineral oil. Amplification conditions were 30 cycles of denaturation at 94°C for 30 s, annealing at 40°C for 30 s, and extension at 72°C for 30 s, final extension at 72°C for 10 min. 5 μl of the product was electrophoresed on a 2% agarose gel. The remaining PCR product was later used for SSCP analyses.

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Two negative controls were included. The 1st was a DNA extraction negative control where extractions were performed without any DNA. The 2nd control was a negative control containing all the substances in the PCR mixture except DNA template. The positive control for the SSCP electrophoresis was obtained by amplifying extracted DNA from 1 *An. funestus* from Tanzania which was chromosomally identified (Green and Hunt 1980). DNA from this specimen was used in every gel throughout the trials.

SSCP Electrophoresis. SSCP electrophoresis was done according to Hiss et al. (1994), with minor alterations. Mutation Detection Enhancement gel solution, 2 times concentration from FMC Bioproducts, Rockland, ME, (catalog number 50620) was diluted to a 0.5-times concentration and used for gel casting. Electrophoresis was performed using a Bio-Rad PROTEIN II xi 2-d Cell (16 by 18 cm) and with 1-mm-thick gels. Glass plates were prepared according to manufacturer's specifications. Different electrophoresis temperatures were tested at 4, 15, and 20°C. The central cooling core helped in keeping temperatures at the correct settings. Before denaturing, 3 μl formamide was added to 10 μl PCR product. The centrifuge tubes (0.5 ml) were put into ice immediately after denaturing and 10 μl of PCR product was electrophoresed for ≈16 h at 200V. The electrode buffer was 0.6 M TBE and a final concentration of TBE in the gel was 0.05 M. A 1Kb-size marker was loaded in the 1st lane on the left-hand side to show the orientation of the gel.

Gels were stained using a general silver staining technique as described in Budowle et al. (1991) and fixed in 10% acetic acid when the desired intensity was reached. Gels were dried in the air or in a heated vacuum drier. Chemicals used for staining, except sodium carbonate and acetic acid, could be reused for up to 3 times without loss of sensitivity.

Results

The morphologically identified species and the number of specimens used in this assay are given in Table 1. All *An. vaneedeni* specimens were identified by the pale banding on the tarsomeres, whereas *An. funestus* had only black scaling on the legs (De Meillon et al. 1977). *Anopheles rivulorum* and *An. leesoni* were identified on larval and egg characteristics (Gillies and DeMeillon 1968).

The D3 primers amplified the same fragment of ≈400 bp, and no obvious size differences were seen among the 4 species when visualized on a 2% agarose gel stained with ethidium bromide. The optimum temperature for electrophoretic separation was at 20°C, and at 4°C, resolution seemed to decrease.

Figure 1 illustrates the species-specific banding patterns observed after SSCP electrophoresis. The upper set of bands represent renatured single-strand (RSS) DNA and the lower set of bands represents the denatured single-strand (DSS) DNA (Hiss et al. 1994). Renatured double-strand DNA, that would have appeared between the RSS and DSS molecules, was never visualized on our gels. For each species, at least

Table 1. Specimens used in SSCP analysis of 4 members of the *An. funestus* group

Country	Locality	No. families	No. individuals	Species
Tanzania	Matimbwa	3	16	*An. funestus*
	Bagamoya	1	1	*An. funestus*
Côte d'Ivoire	Petionara	3	8	*An. funestus*
	Abidja	2	6	*An. funestus*
Senegal	Kedougou	—	6	*An. funestus*
	Sankagne	—	6	*An. funestus*
	Dielmo	—	6	*An. funestus*
Mozambique	Boane	8	26	*An. funestus*
Madagascar	Ankazobe	9	27	*An. funestus*
Botswana	Shakawe	1	6	*An. funestus*
South Africa	Mamfene	1	1	*An. rivulorum*
	Mamfene	10	10	*An. vaneedeni*
	Tzaneen	2	2	*An. leesoni*
	Tzaneen	16	38	*An. vaneedeni*
	Komatipoort	7	30	*An. vaneedeni*
	Komatipoort	3	20	*An. rivulorum*

4 different bands were observed. These are labeled with the 1st letter of each species name (e.g., F = *funestus*, V = *vaneedeni*, etc.), an R or a D to designate renatured or denatured single-strand DNA, and the numbers 1, 2, or 3 to indicate mobility. The RSS bands were used as the diagnostic banding pattern to identify each species. The *An. funestus* F-R1 and F-R2 bands were constant and always present in the 108 specimens from 10 different localities. Intraspecific variation was noted between specimens from East and West Africa, where an additional molecule F-R3 always was absent in the East African samples but present in all of the 14 specimens tested from the 2 localities in Côte d'Ivoire and in 12 of the 18 specimens from Senegal (Fig. 2). The F-R3 molecule was slightly slower than the V-R2 molecule and did not influence identification. The F-R1 molecule was always present, and the F-R3 and F-R1 molecules were half the intensity of F-R2. *An. funestus* from Mozambique had an

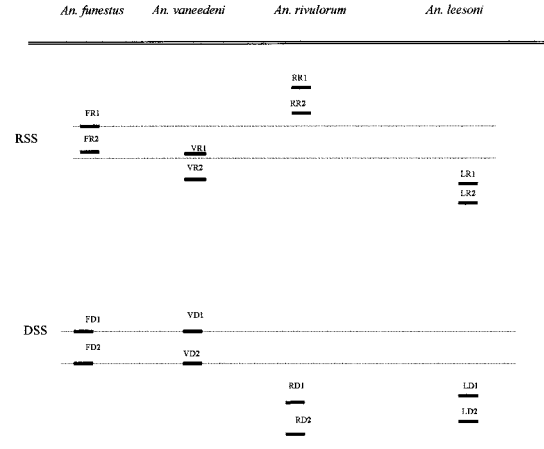


Fig. 1. Species-specific banding patterns observed after SSCP electrophoresis of 4 members of the *An. funestus* group. RSS, renatured single-strand; DSS, denatured single-strand; F, *funestus*, V, *vaneedeni;* R, *rivulorum;* L, *leesoni*.; R1 and R2, RSS molecules; D1 and D2, DSS molecules.

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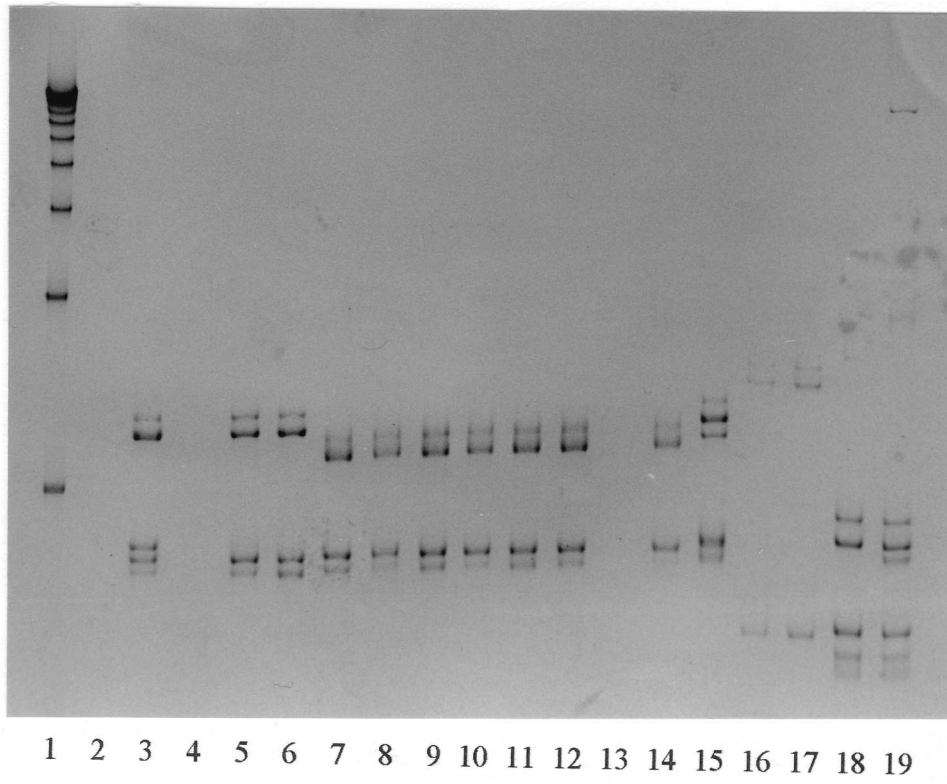


Fig. 2. SSCP analysis showing intraspecific and interspecific variation. Lane 1, 1Kb-size marker; 2, negative control; 3, *An funestus* (Mozambique); 5, *An. funestus* (Tanzania); 6, *An. funestus* (Madagascar); 7–11, *An. vaneedeni* (South Africa, Tzaneen); 12,14, *An. vaneedeni* (South Africa, Komatipoort); 15, *An. funestus* (Côte d'Ivoire, Abidjan); 16–17, *An. rivulorum* (South Africa); 18–19, *An. leesoni* (South Africa). Lanes 4 and 13 did not amplify.

additional F-D molecule (F-D1a) which was half the intensity of the F-D2 molecule and migrated more slowly than F-D1. Other than that, the DSS molecules of *An. funestus* and *An. vaneedeni* migrated at the same speed. The RSS molecules of *An. funestus* from Mozambique did not show intraspecific variation when compared with other *An. funestus* from different regions.

Twenty-one *An. rivulorum* tested showed the same banding patterns and without any variation in either DSS or RSS molecules. Of the 78 *An. vaneedeni* tested, none showed any intraspecific variation, and the banding patterns were constant between runs. Only 2 *An. leesoni* specimens were collected for analysis. Amplification and electrophoresis of these 2 were repeated independently on 8 occasions and the same results were obtained each time. One of the 2 individuals displayed intraspecific variation where L-R3 was present with the L-R1 and L-R2 molecules.

The diagnostic bands for each species are depicted diagrammatically in Fig. 1. Figure 2 shows actual specimens of all 4 species. Migration of the size marker molecules was not constant, and variation was observed when the duration of the electrophoresis run was longer than 16 h.

Discussion

For each single-strand molecule at least 2 different bands will be observed, and for a denatured double-strand molecule at least 4 bands will be visible (Orita et al. 1989). Two of the 4 bands will correspond to the complementary strand and their mobilities will be different because of a different structural folding. Species-specific patterns always were observed with the RSS molecules. Because of the overlap between the DSS molecules of *An. funestus* and *An. vaneedeni*, these sets alone should not be used for species diagnosis. However, both the RSS banding pattern and DSS banding patterns were distinctive for *An. rivulorum* and *An. leesoni*. The RSS banding patterns were stable and no variation between R1 and R2 molecules of the same species were observed except the additional R3 molecule in *An. funestus* specimens from West Africa. This, however, never compromised the usefulness of the SSCPs for identification purposes. There might be sequence differences in West African populations that result in forming an additional conformation. The fact that the F-R1 and F-R3 are both of the same intensity and half the intensity of F-R2 suggest that F-R1 and F-R3 molecules are from the same complementary strand. Their sequence might permit them to form

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either 1 of the 2 conformations and with the same bias. However, the only way to know what causes the difference in banding patterns will be to excise the DNA from the gel and sequence each molecule separately. The banding pattern was constant and present in all the specimens we tested and is not an artifact of the PCR.

The design of species-specific primers involves the accumulating of sequence information on a certain region in the genome of the mosquito. The sequences need to be unique in each species, and should be located in such a manner so that the unique size products can be distinguished easily on agarose gels. The development of primers is an extremely expensive and time-consuming process. We sequenced 2 regions in the rDNA, the D3 domain within the 28S gene and the ITS1 region, but the lack of variation excluded the development of good species-specific primers from these regions (unpublished data). SSCP analysis, however, is a rapid and simple test to perform, the results are not difficult to interpret, and the staining method is straightforward. The primers used for SSCP analysis are generally conserved and can be used for different genera and species.

Acrylamide is less expensive than agarose and casting of polyacrylamide gels (1 mm thick) is not difficult. Mutation detection enhancement gel solution is slightly more expensive than the standard polyacrylamide, but gives excellent resolution. Vertical electrophoretic systems are, however, more expensive, but they are also useful in other studies such as iso-enzyme for population genetic studies. Time needed for electrophoresis is much longer (16 h), but this does not interfere with working hours. SSCP gels can be visualized using radioactivity, fluorescence, or silver staining. Processing radioactive waste is problematic and the danger involved in working with radioactivity puts this method at a disadvantage. Fluorescence must be kept in the dark and the shelf life is short. Silver staining is easy to use and relatively inexpensive. Hiss et al. (1994) showed that silver staining was as sensitive as radioactivity and that even between different electrophoretic conditions, intraspecific and interspecific resolution was the same. Silver staining is more time consuming than staining with ethidium bromide but does not present as severe a biohazard. A transilluminator is not needed for the visualization of the gel. All the chemicals for silver staining are kept at room temperature and can be reused. Only a small amount of each specimen is needed for the PCR assay and the rest of specimen can be used for other tests such as ELISA assays to determine the sporozoite infection rate. Different methods of DNA preservation can be used depending on the purpose and condition (e.g., dried, frozen in nitrogen, preserved in ethanol or other alcohols) and even museum specimens can be identified. This method is not dependent on the presence or absence of a signal and therefore no false positives are possible.

Hiss et al. (1994) were able to distinguish 8 leafhopper species and 5 different mosquito species with similar taxonomic problems. We suspect that even if more species of the *An. funestus* group are incorporated in the study, the banding patterns observed will be different from the species tested so far. We did not expect any variation between males and females because rDNA is sex-linked. We did not have access to hybrid specimens and could not test the impact of hybridization on the technique. Banding patterns possibly will be an overlap of the 2 species which took part in hybridization, especially when looking at females.

We conclude that SSCP analysis is a sensitive and reliable method for identification of the 4 members of the *An. funestus* group.

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